

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Even Semester)

Innovative Practices Description

UNIT I INTRODUCTION

Degree, Semester & Branch: IV Semester B.E. ECE A
Course Code & Title: EC8451 Electromagnetic Fields
Name of the Faculty member: Mrs.S.Jeeva
Name of the Topic: Two Null Identities & Helmholtz's theorem
Name of the Innovative Practice: One Minute Paper
Date & Duration: 06.01.2020 & 5 minutes

Description:

Minute paper is a short writing activity within the classroom in which students generate answers in response to questions asked by the teacher regarding the concepts taught in class. Minute papers can provide a "conceptual bridge" between successive class periods. These questions stimulate a student to plan ahead for the next class.

Goals (Learning Outcomes):

- The students will be able to Apply the basic mathematical concepts of vector analysis.

Use of Appropriate Method:

Justification for choosing the One Minute paper activity for this topic:

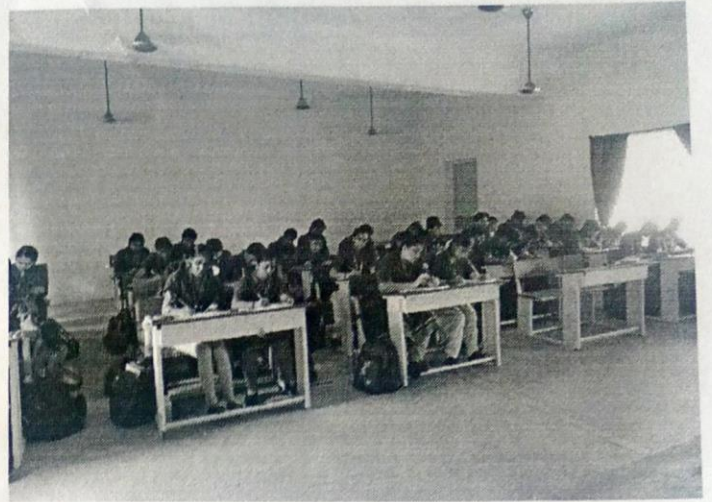
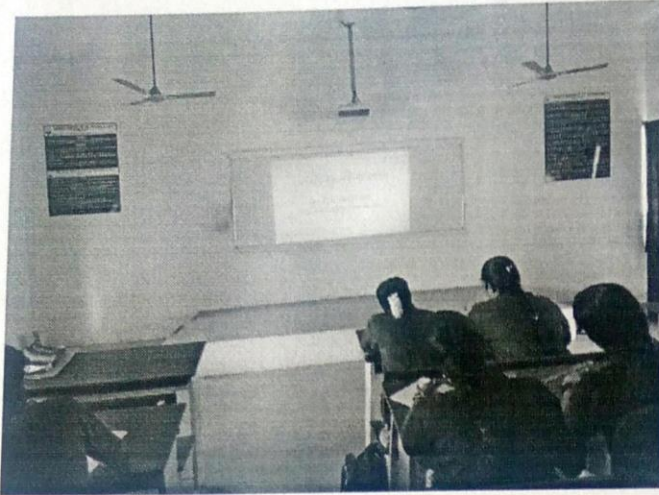
Unit I deals with the basics of vector algebra & vector calculus and theorems which is an important topic in concept wise and university exam point of view. So I have chosen this topic and the reason for choosing mind map activity is while recollecting the concepts and hints learnt in the classroom, the students will be able to reproduce the concept in the exam.

Effective Presentation:

Implementation (Plan & Execution) with Proof:

At the end of the class, students were asked to write answers for the some questions. These questions includes the most important key terms, the students have to learn in that topic.

Execution Images



Sample Minute Papers

N. HANDESHIA
ECE - A
05/06/2020

ACTIVITY : 1 MINUTE PAPER

TOPIC : NULL IDENTITIES, HELMHOLTZ THEOREM

1. NULL IDENTITIES

Identity 1

$$\nabla \times (\nabla \times \mathbf{V}) = \nabla(\nabla \cdot \mathbf{V}) - \nabla^2 \mathbf{V}$$

Identity 2

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0$$

2. HELMHOLTZ THEOREM

Let $\mathbf{F}$ be a vector field. Then $\mathbf{F}$ can be decomposed into a solenoidal part and an irrotational part.

Solenoidal part: $\nabla \cdot \mathbf{F} \neq 0, \nabla \times \mathbf{F} = 0$

Irrotational part: $\nabla \cdot \mathbf{F} = 0, \nabla \times \mathbf{F} \neq 0$

Neither solenoidal nor irrotational: $\nabla \cdot \mathbf{F} \neq 0, \nabla \times \mathbf{F} \neq 0$

N. HANDESHIA
ECE - A
05/06/2020

ACTIVITY : 1 MINUTE PAPER

TOPIC : NULL IDENTITIES, HELMHOLTZ THEOREM

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N. HANDESHIA
ECE - A
05/06/2020

ACTIVITY : 1 MINUTE PAPER

TOPIC : NULL IDENTITIES, HELMHOLTZ THEOREM

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2. HELMHOLTZ THEOREM

Let $\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 \rightarrow$ solenoidal & irrotational

(a) Solenoidal & irrotational
 (b) Solenoidal but not irrotational
 (c) Irrotational but not solenoidal
 (d) Neither solenoidal nor irrotational

N. HANDESHIA
ECE - A
05/06/2020

ACTIVITY : 1 MINUTE PAPER

TOPIC : NULL IDENTITIES, HELMHOLTZ THEOREM

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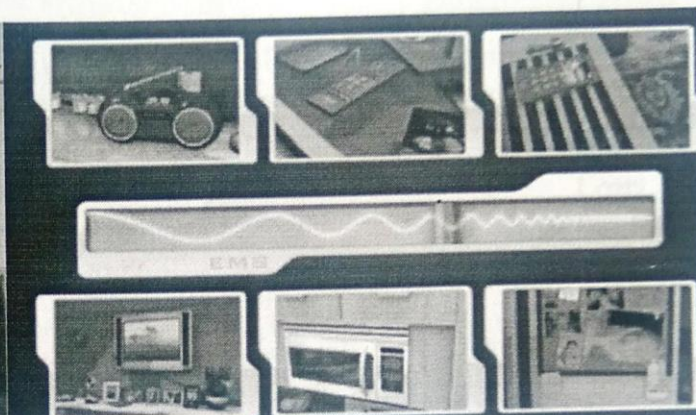
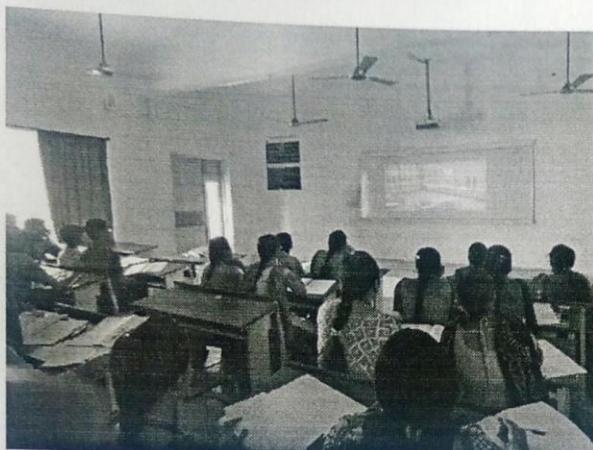
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Name of the Topic: Electromagnetic Model (Electromagnetic Spectrum)

Name of the Innovative Practice: Video Clip

Date & Duration: 16.12.2019 & 10 minutes

Execution Images

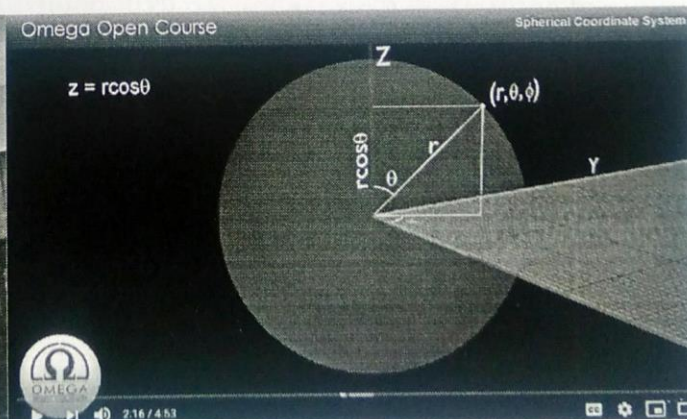


Name of the Topic: Spherical coordinate system

Name of the Innovative Practice: Animated video

Date & Duration: 19.12.2019 & 10 minutes

Execution Images



Significance of Results:

Assessment of effectiveness/Success of the activity:

By conducting this activity, the students were able to recall the topics.

Reflective Critique:

Benefits:

Minute papers can improve the quality of class discussion by having students write briefly about a concept or issue before they begin discussing it.

Challenges:

As it is mentioned above that it takes one minute to conduct this activity. But, the students are failed to remember all equations in a given duration. They took five minutes to complete this activity.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	-	-	-	2	-	2	-	1

CO1: The students will be able to Apply the basic mathematical concepts of vector analysis.

References:

- (i) M.N.O. Sadiku and S.V. Kulkarni, Principles of electromagnetics, 6th ed., Oxford (Asian Edition), 2015
- (ii) D.K. Cheng, Field and wave electromagnetics, 2nd ed., Pearson (India), 1989.
- (iii) <https://www.youtube.com/watch?v=cfXzwh3KadE>
- (iv) <https://www.youtube.com/watch?v=FDyenWWIPdU>